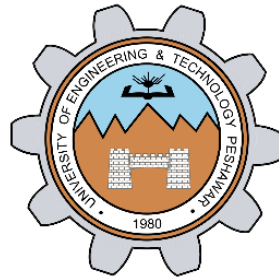


Object Oriented Programming Lab

Revision of Structured Programming

LAB # 00



Fall 2025

CSE208L Object Oriented Programming Lab

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Registration No.: **24PWCSE2382**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Student Signature: _____

Submitted to:

Engr. Sumayyah Salahuddin

September 22, 2025

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Object Oriented Programming Lab

Objectives of the Lab:

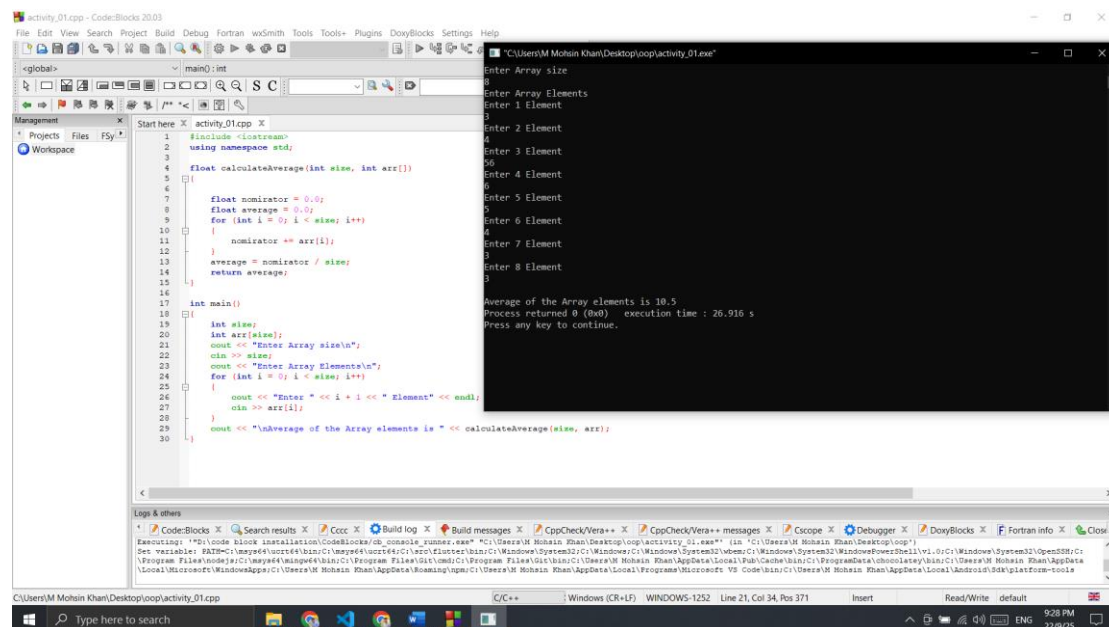
- Reinststate the concepts of structured programming such as:
 - Decision making
 - Iterations
 - Functions
 - Structures and arrays
 - Pointers

Activity # 0.3.1

Title:

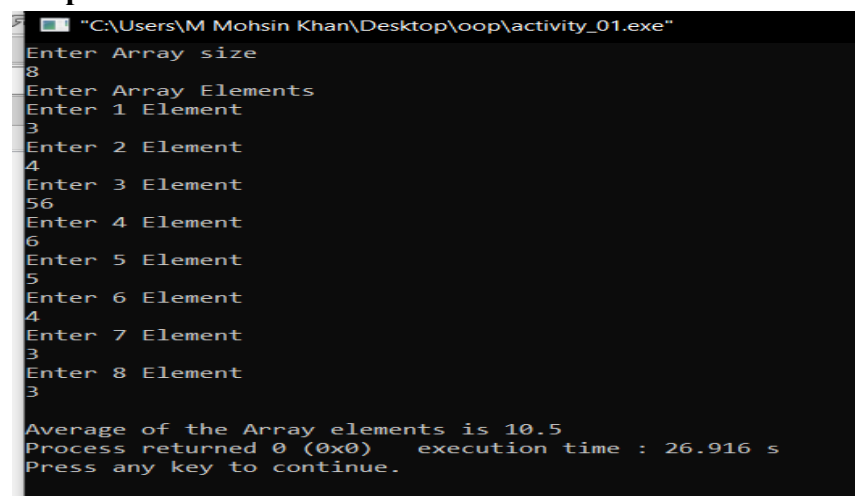
Write a C++ program that calculates the average of a set of numbers entered by the user.

Source Code:



```
1 #include <iostream>
2 using namespace std;
3 float calculateAverage(int size, int arr[])
4 {
5     float nominator = 0.0;
6     float average = 0.0;
7     for (int i = 0; i < size; i++)
8     {
9         nominator += arr[i];
10    }
11    average = nominator / size;
12    return average;
13 }
14
15 int main()
16 {
17     int size;
18     int arr[size];
19     cout << "Enter Array size\n";
20     cin >> size;
21     cout << "Enter Array Elements\n";
22     for (int i = 0; i < size; i++)
23     {
24         cout << "Enter " << i + 1 << " Element" << endl;
25         cin >> arr[i];
26     }
27     cout << "Average of the Array elements is " << calculateAverage(size, arr);
28 }
```

Output:



```
"C:\Users\M Mohsin Khan\Desktop\oop\activity_01.exe"
Enter Array size
8
Enter Array Elements
Enter 1 Element
3
Enter 2 Element
4
Enter 3 Element
56
Enter 4 Element
6
Enter 5 Element
5
Enter 6 Element
4
Enter 7 Element
3
Enter 8 Element
3

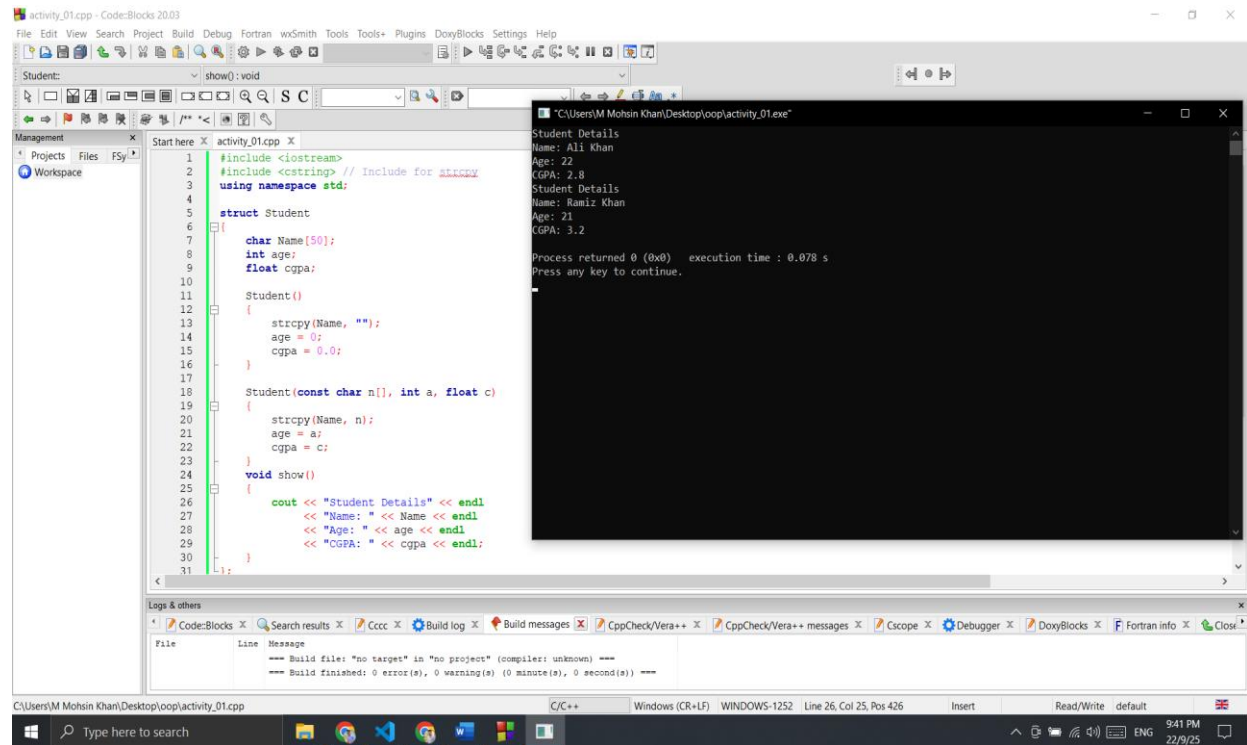
Average of the Array elements is 10.5
Process returned 0 (0x0)   execution time : 26.916 s
Press any key to continue.
```

Object Oriented Programming Lab

Activity # 0.3.2

Write a program that creates Student structure.

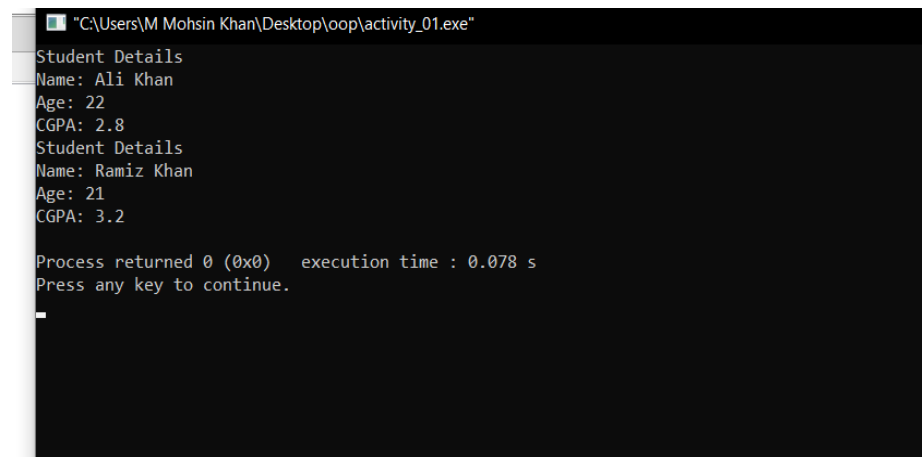
Source Code:



```
1 #include <iostream>
2 #include <string> // Include for strcpy
3 using namespace std;
4
5 struct Student
6 {
7     char Name[50];
8     int age;
9     float cgpa;
10
11     Student()
12     {
13         strcpy(Name, "");
14         age = 0;
15         cgpa = 0.0;
16     }
17
18     Student(const char n[], int a, float c)
19     {
20         strcpy(Name, n);
21         age = a;
22         cgpa = c;
23     }
24     void show()
25     {
26         cout << "Student Details" << endl;
27         << "Name: " << Name << endl;
28         << "Age: " << age << endl;
29         << "CGPA: " << cgpa << endl;
30     }
31 };
32
33 int main()
34 {
35     Student s1;
36     Student s2("Ali Khan", 22, 2.8);
37     s1.show();
38     s2.show();
39     return 0;
40 }
```

Student Details
Name: Ali Khan
Age: 22
CGPA: 2.8
Student Details
Name: Ramiz Khan
Age: 21
CGPA: 3.2
Process returned 0 (0x0) execution time : 0.078 s
Press any key to continue.

Output:



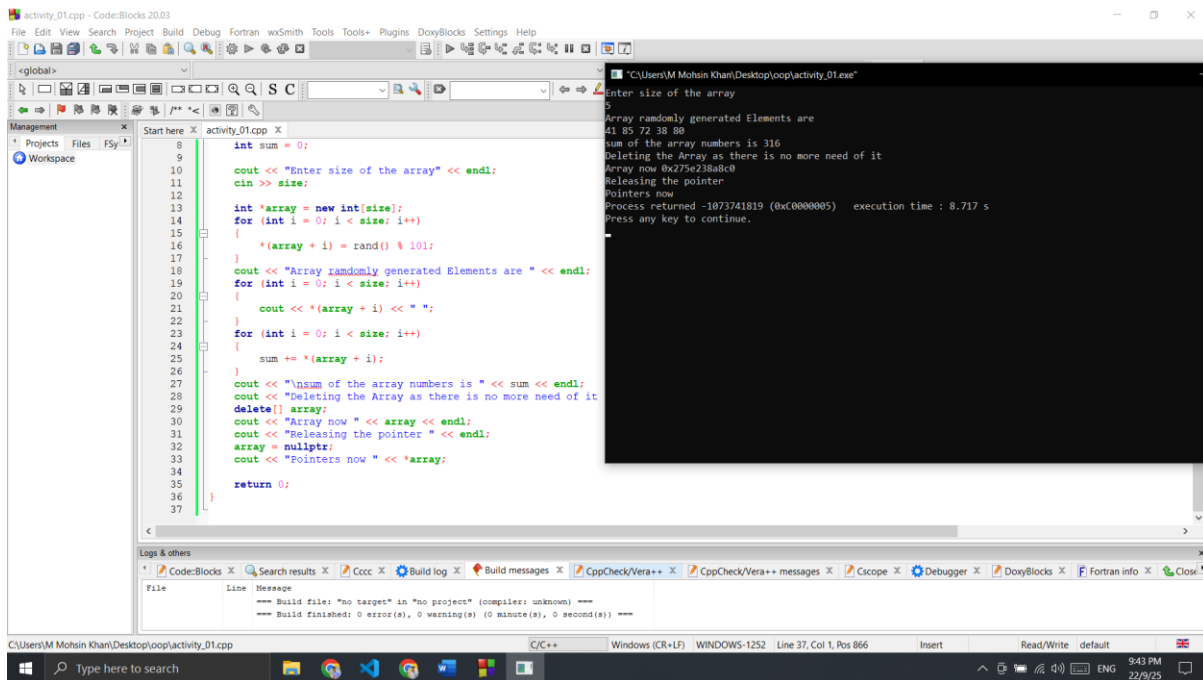
```
"C:\Users\M Mohsin Khan\Desktop\oop\activity_01.exe"
Student Details
Name: Ali Khan
Age: 22
CGPA: 2.8
Student Details
Name: Ramiz Khan
Age: 21
CGPA: 3.2
Process returned 0 (0x0) execution time : 0.078 s
Press any key to continue.
```

Object Oriented Programming Lab

Activity # 0.3.3

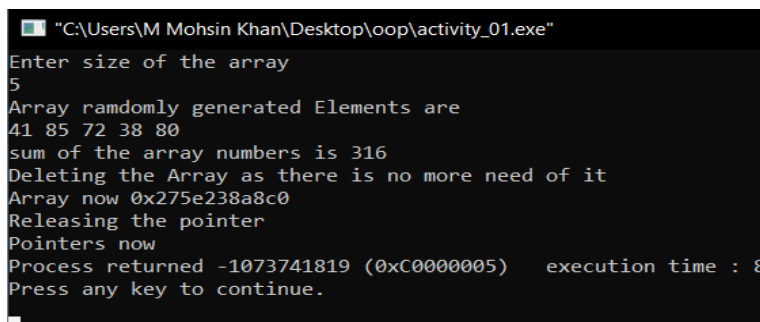
Write a C++ program that generates random data and calculates the sum of the generated data.

Source Code:

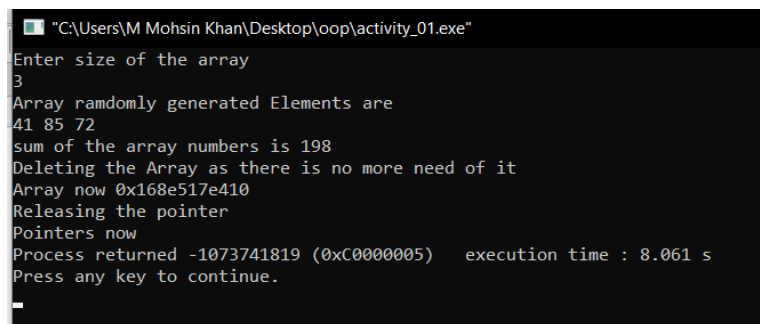


```
activity_01.cpp - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
<global>
Management
Projects Files FSy
Workspace
Start here X activity_01.cpp X
8 int sum = 0;
9
10 cout << "Enter size of the array" << endl;
11 cin >> size;
12
13 int *array = new int[size];
14 for (int i = 0; i < size; i++)
15 {
16     *(array + i) = rand() % 101;
17 }
18 cout << "Array randomly generated Elements are " << endl;
19 for (int i = 0; i < size; i++)
20 {
21     cout << *(array + i) << " ";
22 }
23 for (int i = 0; i < size; i++)
24 {
25     sum += *(array + i);
26 }
27 cout << "sum of the array numbers is " << sum << endl;
28 cout << "Deleting the Array as there is no more need of it" << endl;
29 delete[] array;
30 cout << "Array now " << array << endl;
31 cout << "Releasing the pointer " << endl;
32 array = nullptr;
33 cout << "Pointers now " << array;
34
35 return 0;
36
37
Logs & others
Code::Blocks X Search results X Cccc X Build log X Build messages X CppCheck/Ver++ X CppCheck/Ver++ messages X Cscope X Debugger X DoxyBlocks X Fortran info X Close
File Line Message
=== Build file: "no target" in "no project" (compiler: unknown) ===
=== Build finished: 0 error(s), 0 warning(s), 0 minute(s), 0 second(s) ===
C:\Users\M Mohsin Khan\Desktop\oop\activity_01.cpp C/C++ Windows (CR+LF) WINDOWS-1252 Line 37, Col 1, Pos 866 Insert Read/Write default
Type here to search 9:43 PM 22/9/25
```

Output:



```
"C:\Users\M Mohsin Khan\Desktop\oop\activity_01.exe"
Enter size of the array
5
Array randomly generated Elements are
41 85 72 38 80
sum of the array numbers is 316
Deleting the Array as there is no more need of it
Array now 0x275e238a8c0
Releasing the pointer
Pointers now
Process returned -1073741819 (0xC0000005) execution time : 8
Press any key to continue.
```



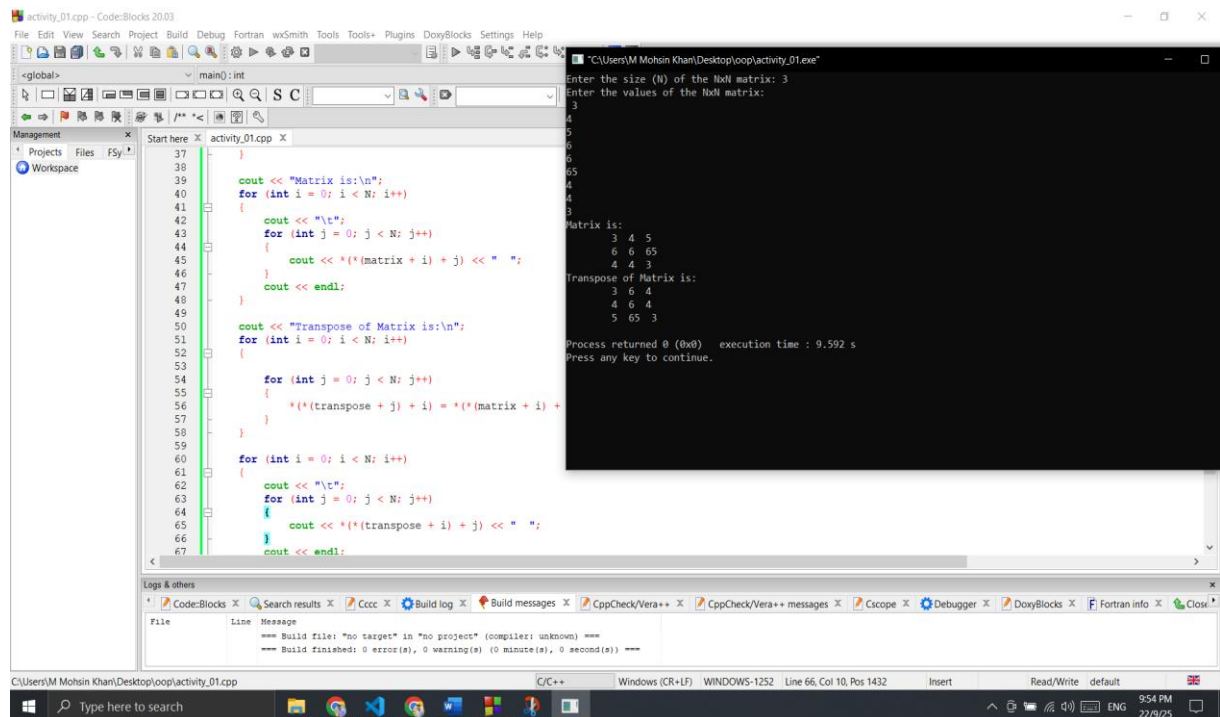
```
"C:\Users\M Mohsin Khan\Desktop\oop\activity_01.exe"
Enter size of the array
3
Array randomly generated Elements are
41 85 72
sum of the array numbers is 198
Deleting the Array as there is no more need of it
Array now 0x168e517e410
Releasing the pointer
Pointers now
Process returned -1073741819 (0xC0000005) execution time : 8.061 s
Press any key to continue.
```

Object Oriented Programming Lab

Activity # 0.3.4

Write a C++ program to find transpose of a NxN matrix. Declare integer matrix and its transpose using calloc () (use #include <cstdlib>). Use 2D pointers in this activity for storage, retrieval, and finding transpose. Free the pointer variables in the end.

Source Code:

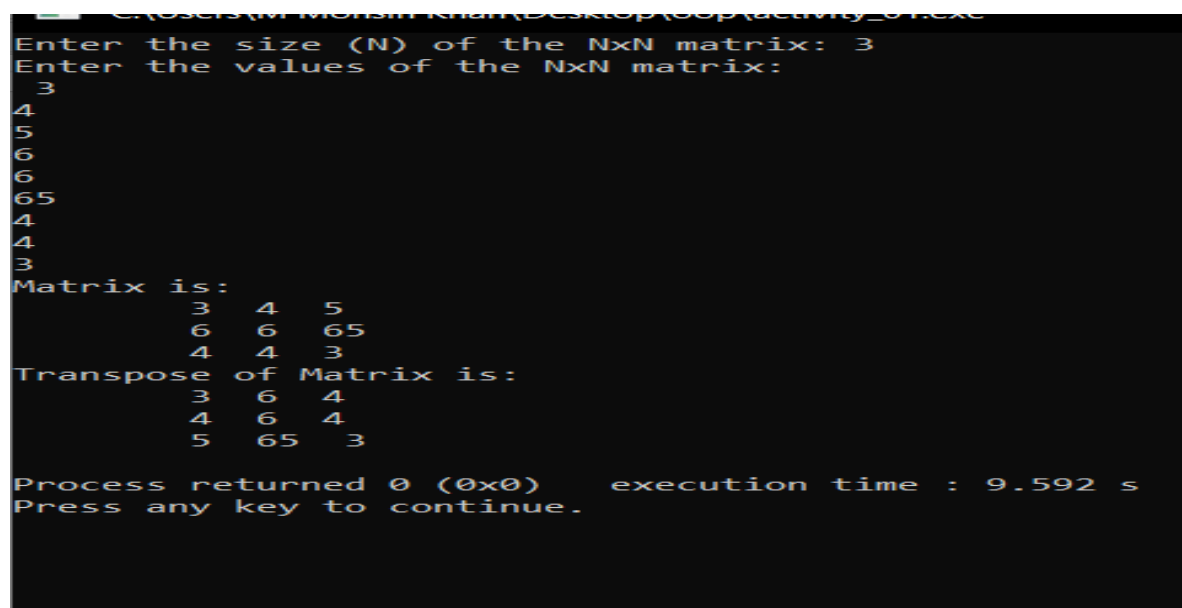


```
activity_01.cpp - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
<global>
main0: int
37
38
39 cout << "Matrix is:\n";
40 for (int i = 0; i < N; i++)
41 {
42     cout << "\t";
43     for (int j = 0; j < N; j++)
44     {
45         cout << *((matrix + i) + j) << " ";
46     }
47     cout << endl;
48 }
49
50 cout << "Transpose of Matrix is:\n";
51 for (int i = 0; i < N; i++)
52 {
53     for (int j = 0; j < N; j++)
54     {
55         *((transpose + j) + i) = *((matrix + i) + j);
56     }
57 }
58
59 for (int i = 0; i < N; i++)
60 {
61     cout << "\t";
62     for (int j = 0; j < N; j++)
63     {
64         cout << *((transpose + i) + j) << " ";
65     }
66     cout << endl;
67 }
```

Output:

```
Enter the size (N) of the NxN matrix: 3
Enter the values of the NxN matrix:
3
4
5
6
65
4
4
3
Matrix is:
3 4 5
6 6 65
4 4 3
Transpose of Matrix is:
3 6 4
4 6 4
5 65 3
Process returned 0 (0x0) execution time : 9.592 s
Press any key to continue.
```

Output:



```
Enter the size (N) of the NxN matrix: 3
Enter the values of the NxN matrix:
3
4
5
6
65
4
4
3
Matrix is:
3 4 5
6 6 65
4 4 3
Transpose of Matrix is:
3 6 4
4 6 4
5 65 3
Process returned 0 (0x0) execution time : 9.592 s
Press any key to continue.
```